Segmentation fault when using VQE with COBYLA optimizer

https://github.com/qiskit-community/qiskit-aqua/issues/80

ROW 64

Segmentation fault when using VQE with COBYLA optimizer #80

Closed

yaelbh opened on Aug 9, 2018

Informations

- **Qiskit Aqua version**: current master
- **Python version**: 3.6
- **Operating system**: Linux

What is the current behavior?

A segmentation fault occurs for a specific input.

Steps to reproduce the problem

Create the following input file of aqua chemistry:

```
=====================================================

&name
LiH excited states molecule experiment. Var for SWAPRZ with VQKE
&end

&driver
name=PYSCF
&end

&problem
name=energy
enable_substitutions=True
random_seed=50
&end

&pyscf
atom=H 0 0 -2.5; Be 0 0 0; H 0 0 2.5
unit=Angstrom
charge=0
spin=0
basis=sto3g
&end

&operator
name=hamiltonian
transformation=particle_hole
qubit_mapping=jordan_wigner
two_qubit_reduction=False
freeze_core=True
orbital_reduction=[]
max_workers=4
&end

&algorithm
name=VQE
operator_mode=matrix
initial_point=None
&end

&optimizer
name=COBYLA
maxiter=10000
&end

&variational_form
name=SWAPRZ
depth=1000
entanglement=full
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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&variational_form
name=SWAPRZ
depth=1000
entanglement=full
entangler_map=None
&end

&initial_state
name=HartreeFock
qubit_mapping=jordan_wigner
two_qubit_reduction=False
&end

&backend
name=local_statevector_simulator
&end

==

Then run:
qiskit_aqua_chemistry_cmd <file name>

The input cam from Julia Rice, but with other values of depth and maxIter. The bug occurs for these specific depth and maxIter (depth=1000, maxIter=10000). I haven't witnessed it for other values of these parameters.

What is the expected behavior?

Run without segmentation fault.

Suggested solutions

For the moment I only know that it happens inside COBYLA.optimize.

yaelbh on Aug 9, 2018 Contributor Author ···

After the call to scipy.optimize, the segmentation fault happens before the call to vqe._energy_evaluation

chunfuchen on Aug 9, 2018 Contributor ···

I think this is the problem of COBYLA optimizer. With `depth=1000`, you will have 78000 parameters.
It seems that it exceeds what COBYLA can do.

Here is a simple test about COBYLA (not involving aqua)

```
from scipy.optimize import rosen, minimize
import numpy as np

x0 = np.random.random(78000) # number of paramters
res = minimize(rosen, x0, tol=1e-06, method="COBYLA")
```

It raises segmentation error as well.

I suggest to reduce the depth size or change the optimzer.

yaelbh on Aug 12, 2018 Contributor Author ···

Great, thanks, closing the issue then.

yaelbh closed this as completed on Aug 12, 2018